Tomato, scientifically known as *Lycopersicum esculentum Miller* is believed to have originated from South America and was brought by the Spaniards to Europe in 1500s and later to the Philippines which spread throughout Asia. It is an important and popular vegetable grown in all parts of the country. It ranks second to eggplant in terms of total production area. It is regarded as one of the most profitable crops for off-season production, preferably from May to September.

Tomato is the top source of Vitamins A and C. It also contains significant amount of dietary fiber, beta carotene, iron lycopene, magnesium, niacin, potassium, phosphorous, riboflavin and thiamine. Tomato is rich in lycopene, an anti-oxidant which destroys cancer-causing free radicals in the body. It is also good for the heart, helps prevent premature aging and reduces the risks of chronic diseases.

Fresh tomatoes are used in many ways, such as:

- Ingredient in many recipes
- Served as raw, baked, stewed and as sauce for other foods

There are two types of tomatoes grown in the country namely, the table type and the processing type. Table type tomatoes are non-seasonal while the processing type tomatoes are grown only during dry season.

SOIL AND CLIMATIC REQUIREMENTS

Tomato requires a relatively cool, dry climate for high yield and premium quality. It is adapted to a wide range of climatic conditions. The optimum temperature requirement is 21 to 24° C. Plant tomato from September to January in hilly areas and from November to February in lowland areas. Grow off-season type from May to September for bigger profit.

Tomatoes can be grown in varied soil types such as sandy loam to clay-loam with rich organic matter but should not be planted in areas with long period of flooding. The ideal soil pH is 5.5 - 8.0.

Choose a part of the farm that is slightly elevated with good drainage to avoid water logging in case of flash flood during the wet season. For dry season planting, make a catchment canal that will drain the excess water after irrigation.

CULTURAL MANAGEMENT PRACTICES

Selection of Varieties

Yield potential, adaptability and marketability of tomato varieties determine the income of growers. Hence, selection of variety should be according to this purpose. To guide you in the selection of appropriate variety to plant refer to the *Guide in Selecting Lowland Vegetable Varieties* included in this kit.

Land Preparation

Prepare the field as early as possible to give enough time for the weeds and stubbles of previous crops to decompose. Plow and harrow 2 to 3 times alternately at one week interval. Plow at a depth of 15 to 20 cm. Harrow twice to break the clods and level the field. A well-pulverized soil promotes good soil aeration and enhances root formation.

Raising Seedlings

A hectare of farm requires 150 to 200 grams of seeds.

There are two methods of rearing seedlings: the seedbed method and the seedling trays or potlets method.

Seedbed Method

- Choose a level area fully exposed to sunlight, accessible to water source, with good drainage and provided with windbreaks.

- Prepare the area by plowing and harrowing alternately until the soil is reduced to fine tilth.
- For one hectare area to be planted, construct seedbed measuring 1x10 m at 15 cm high.
- Pulverize the soil and level the bed with the use of hand tools and remove weeds and stubbles.
- Before sowing the seeds, sterilize the seedbed to kill weed seeds and pathogens present in the soil.
    - Water the prepared seedbed first for better heat penetration.
    - Spread about 3 to 5 cm thick of either rice straw or rice hull on top of the seedbeds and burn slowly.
    - When the soil cools, remove unburned materials and excess ash.
    - To avoid toxic effects of burned materials, do not sow seeds immediately after sterilization.
    - Water the seedbeds before sowing.
    - The plots can also be drenched with boiling water to sterilize.
- The night before sowing, soak the required amount of seeds in clean water. After soaking, sow immediately or air dry the seeds before sowing. Sow the seeds in small shallow furrows at a rate of 20 g per square meter. Cover the seeds slightly with fine soil.

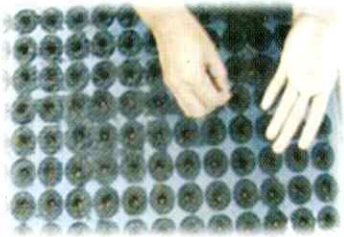
- Dust the surroundings of the seedbed with insecticide powder to avoid ants, and spray 1 tbsp of fungicide per gallon of water to avoid infection of damping-off disease.
- To ensure uniform germination of the seeds, saturate the seedbed with water for the first three days using sprinkler until the seeds germinate.
- Regulate watering as soon as the seedlings emerged to avoid soft and succulent stems.
- Drench the seedlings with a starter solution of urea at a rate of 1 tbsp/gal of water at 7 to 14 days after emergence (DAE). Sprinkle water on the seedlings using a sprinkler immediately after applying fertilizer to avoid burning effect on the leaves.
- Drench the seedbeds with recommended fungicide at the rate of 1 tbsp/gal of water when damping-off is observed.
- One week after transplanting, harden the seedlings by gradually withholding water and exposing them to direct sunlight. Hardening is done to prepare the seedlings to direct field condition.

Seedling Tray Method

- Prepare a potting medium by mixing thoroughly 1:1:1 garden soil, carbonized rice hull and compost or organic fertilizer.
- Fill the potlets or seedling trays with mixed media.
- Sterilize by drenching with boiling water.
- When the mixture is cooled, drop 2 to3 grains of complete fertilizer (14-14-14) in each tray hole before filling with the soil mixture. Fill the holes of the tray/potlets with the mixture and slightly compact it using your palm.

- The night before sowing, soak the required amount of seeds in clean water. After soaking, plant immediately or air dry the seeds before sowing.

- Plant one seed in hole at a depth of 0.5 cm. cover the seeds with fine soil. Seedlings (with 2 leaves) from the seedbed can also be pricked to the seedling trays.



- Water the newly planted seeds or newly pricked seedlings using sprinkler with fine droplets.

Care of Seedlings

Protect the seedlings from excessive sunlight and rain by providing a temporary shade using available indigenous materials such as plastic, cogon, talahib or coconut leaves. Construct the shade to 120 cm high on the east side and about 60 cm high on the west side.

Water the seedlings preferably in the morning to minimize damping off.
The amount of water to be applied should be just enough to keep the soil moist. Over-watering favors damping-off and production of weak seedlings.

Drench the seedlings with starter solution of one tbsp urea (45-0-0) dissolved in one gal of water 10 days after emergence. Water the seedlings immediately after applying fertilizer to prevent leaf injury.

One week before transplanting, harden the seedlings by gradually exposing them to sunlight, and withholding water.

Construction of Planting Beds and Furrows

(a) *When using plastic mulch*, construct 1 m wide beds raised at 20 cm height. Space the beds 0.5 m apart which serve as path for manual watering and furrow irrigation.

(b) *For the conventional method*, set furrows (dry months) or ridges (rainy months) 1 m apart.

Mulching

Mulching controls weeds, conserves soil moisture, prevents soil erosion and leaching of fertilizers, and reflects sunlight to repel insect pests hiding under the leaves.

Mulches such as coconut choir dust, rice straw, rice hull and plastic mulch can be used. For coconut choir dust, rice straw and rice hull mulches, apply after hilling-up with at least 3 to 5 cm thick. Plastic mulch must be applied before transplanting.

Placing the plastic mulch:

(1) Stretch the plastic mulch over the planting bed, with the silver color on top.

(2) Fasten the edges of the plastic mulch to the soil with bamboo slats spaced 30 cm apart (6 to 8 cm long).

If bamboo slats are not available, cover its edges with soil using a shovel or rake, or by passing a moldboard plow near the edges.

Making the planting holes:

(1) Measure 0.5 m planting distance 10 cm from the edge on both sides of the plastic mulch and mark the spots.

(2) Make the holes following any of these three ways:

- *Using a serrated-lipped tin can* - Cut the lip of a 7 to 10 cm diameter tin can with metal scissor to form 1-cm long saw-like edges. Push the can on the marked planting spots of the plastic sheet, serrated side down.

- *Using a hot tin can* - Punch small holes using a nail at the sides and bottom of a 7 to10 cm diameter tin can. Attach a wooden or bamboo handle. Fill the can with burning charcoal. The holes ventilate the charcoal to keep it burning. Push the tin can with the heated bottom on the marked planting spots of the plastic sheets.

- *Using a cutter* - Cut intersecting lines at 7 to10 cm long to form "+" or "X" at the marked planting spots. During transplanting, fold up the 4 flaps to form square planting holes.

Transplanting

For wet season planting, use one month old seedlings because these are harder, taller, and can withstand the impact of rain. Transplant seedlings at a space of 0.5 m between hills during dry season and 0.75 m during wet season. For dry season, transplant 25 day-old seedlings. Transplant right after furrow irrigation.

To avoid breaking the stem of seedlings during transplanting, hold the roots with the thumb and forefinger then push towards the soil at 3 to 5 cm deep depending on the length of the stem. For seedlings in trays, transplant together with the soil from the tray hole.

Transplant late in the afternoon or during cloudy days. Cover newly planted seedlings with banana stalk or any available material for covering.

Fertilizer Application

The quantity and kind of fertilizer to be applied depend on the soil fertility and on the previous cropping patterns based on the result of the soil analysis. In the absence of soil analysis, use the following recommended rates:

Time of application	*Kind of fertilizer*	*Qty of fertilizer*	*Stage of the crop*
Before land preparation	Decomposed animal manure or processed organic fertilizer	5,000 kg/ha	
Basal	Complete (14-14-14)	1 tb per plant	Transplanting time
Starter solution	Urea (45-0-0)	1 tb per gal of water	Seedling at nursery

Time of application	Kind of fertilizer	Qty of fertilizer	Stage of the crop
Sidedress	Urea (46-0-0) Muriate of Potash (0-0-60)	15g per plant 15g per plant	10-15 days after transplanting (DAT)
	Urea (46-0-0) Muriate of Potash (0-0-60)	15g /plant 15g/plant	30 (DAT)

Basal fertilizer is important to achieve a healthy and vigorous seedlings before flowering. Delayed application will result in weaker plants and smaller fruits.

Trellising

Support the transplanted seedlings with trellis at 15 days after transplanting using locally available materials.

Weeding and Cultivation

Cultivate in between rows of plants by off -barring at 14 to 21(DAT). Hill-up at 28 to 35 DAT. Spot-weed just after off- barring and hilling-up if there are standing weeds.

Irrigation

Irrigate the plants as the need arises. Irrigate the field four to five times from transplanting to 14 days before the last harvest depending on the type of soil. Tomato is very sensitive to flooding, hence, irrigation must be done just to moisten the root zone especially during the onset of flowering up to the last harvest. Irrigate by furrow (quick passing) to minimize soil erosion and to favor high fruit setting.

In the absence of surface irrigation, hand water the plants weekly at the rate of one liter per hill until two weeks before the last harvest.

Pruning

Pruning is done to attain better plant growth, uniform and earlier harvest of fruits. It can be done at 10 to 20 days DAT or when the side branches start to elongate. Prune all side branches below the split leaving one branch nearest to the split. Remove also the bud growth on the split and old leaves. Prune in the morning to hasten healing of plant wounds.

Crop Protection

Insect pests and diseases decrease the yield of tomato if not managed properly. Common pests of tomato are thrips, whiteflies and fruitworm. Most prevalent diseases are mosaic virus bacterial wilt and nematodes infection. These pests and diseases can be managed by using resistant varieties, cultural, biological, and chemical control measures *(Refer to Pests and Diseases Management Guide included in this kit).*

Harvesting

Harvesting starts at 55 to 65 DAT. Tomato extend its growth to more or less 100 DAT depending on the variety and management practices employed. First priming can start 50 to 60 (DAT) or 15 to 20 days from flowering stage.

Harvest at weekly interval depending on market demand. Tomatoes can be harvested at three stages, namely:

> Mature green - harvest when fruits start to show green/cream at the blossom end. This is applicable for distant market.

> Pink breaker - harvest when the blossom end turn pinkish or reddish. This is applicable for nearby market.

> Red ripe - harvested for home consumption.

Postharvest

Sorting

- Sort according to size, color and shape.
- Discard damaged fruits.
- Wash with clean water and wipe with clean, dry and soft cloth.

Packing

- Pack fruits in suitable containers that are good and strong enough to protect the commodity.

- Use bamboo containers (kaing or basket), wooden crates, food grade plastic crates, boxes or polyethylene bags.
- Line the bottom of the containers with old newspapers or any cushioning materials.
- Provide ventilation space to allow free air circulation.
- In using polyethylene bags, put two pinholes per kg of fruits to preserve freshness for as long as 10 to 14 days after harvest.